

● WORKING DEMO BUILD

NER Smart Logistics & Accessibility Intelligence

Multimodal route planning, live landslide and flood risk, and settlement-level accessibility scoring for the eight North Eastern states.

PROBLEM STATEMENT

SIH26002

MINISTRY

MDoNER

THEME

Smart Automation

TEAM

CodeCrew

Smart India Hackathon 2026 · Software · Built and running end to end

Eight states, one of the hardest logistics geographies in India

The LEADS 2025 report puts North Eastern states across the full performance range - Mizoram an Exemplar, while Nagaland, Arunachal Pradesh, Manipur and Assam still trail. The causes are structural, not administrative: hill terrain, monsoon blockages, single-lifeline corridors, and transport networks that do not connect to each other.



Single points of failure

NH-10 is Sikkim's only road lifeline. NH-2 is Manipur's. One landslide isolates a state capital.



Monsoon-driven blockage

Corridors close on antecedent rainfall, not on the day's downpour - so closures are predictable, and are not predicted.



Modes that do not co-plan

Road, rail, Brahmaputra waterway and UDAN air cargo are planned in separate silos with no shared view.



Access is never scored

Nothing tells you a bridge cannot take the truck, or that the last 18 km has no motorable road at all.

Five surfaces over one shared engine



Route Planner

Ranked multimodal routings with a grounded explanation of why the top one won.



Network Risk

Live corridor risk, condition-derived alerts, and who gets stranded if corridors close.



Accessibility Atlas

0-100 index for every settlement over six auditable sub-scores.



Field Mode

The same plan as sub-160-character SMS / USSD pages for 2G districts.



Model & Data

What the model is, how well it does, and which data is real.

One engine underneath

SETTLEMENTS

59

8 NE states + Siliguri and Kolkata gateways

CORRIDORS

114

Road, rail, waterway and air cargo

NETWORK MODELLED

17,712 km

Real NH / NFR / NW-2 / UDAN topology

LIVE WEATHER NODES

59

Open-Meteo, refreshed every 15 minutes

CARGO PROFILES

6

Each genuinely changes the chosen route

STRESS SCENARIOS

6

Re-run through the same models

SCOPE

What this demo is - and what it is not

A demo that blurs this line is worth less than one that does not. Everything below is stated the same way in the repository README and on the product's own Model & Data screen.



Real and running

- Settlement coordinates, elevations and states for 59 places
- Corridor topology: actual NH numbers, the NF Railway network, NW-2 Brahmaputra, NW-16 Barak, scheduled and UDAN-RCS air sectors
- Live rainfall for all 59 nodes from Open-Meteo, every 15 minutes
- Routing, risk, accessibility and explanation engines - all computed, none scripted
- A classifier genuinely trained at every server start



Demo data layer

- Per-corridor landslide and flood susceptibility priors
- Pavement condition, bridge load class, gradient, night-travel safety
- Settlement accessibility attributes: ramp, step-free, assisted boarding, last-mile gap, nearest health facility, coverage
- The historical disruption record the classifier learns from - synthesised from monsoon climatology, because open per-corridor incident data is not downloadable

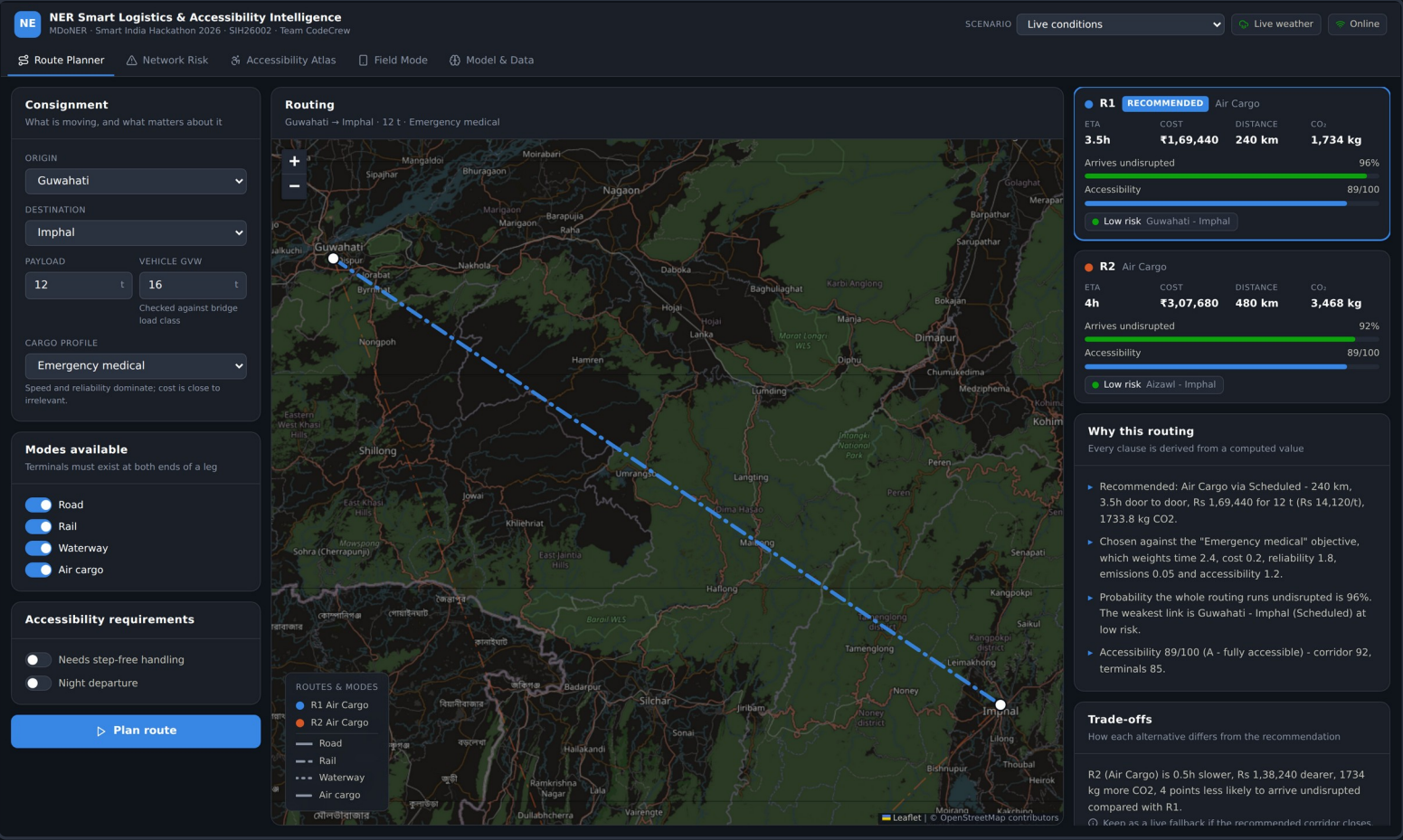


Not built

- Reinforcement learning. The planner is a deterministic multi-objective shortest path - calling that RL would be a lie
- Live sensor, satellite or vehicle-telematics ingestion
- A language model. Explanations come from a grounded template engine, so the system cannot state what the data does not support
- Authentication, multi-tenancy, and any production hardening

Route Planner

Plans a consignment across the multimodal network and shows the reasoning, not just the answer.



1 Ranked routings

Three genuinely distinct options from Yen's k-shortest-paths, not one route relabelled.

2 Grounded explanation

Every clause cites a value the planner computed - distance, weakest link, rainfall, both risk models.

3 Live constraints

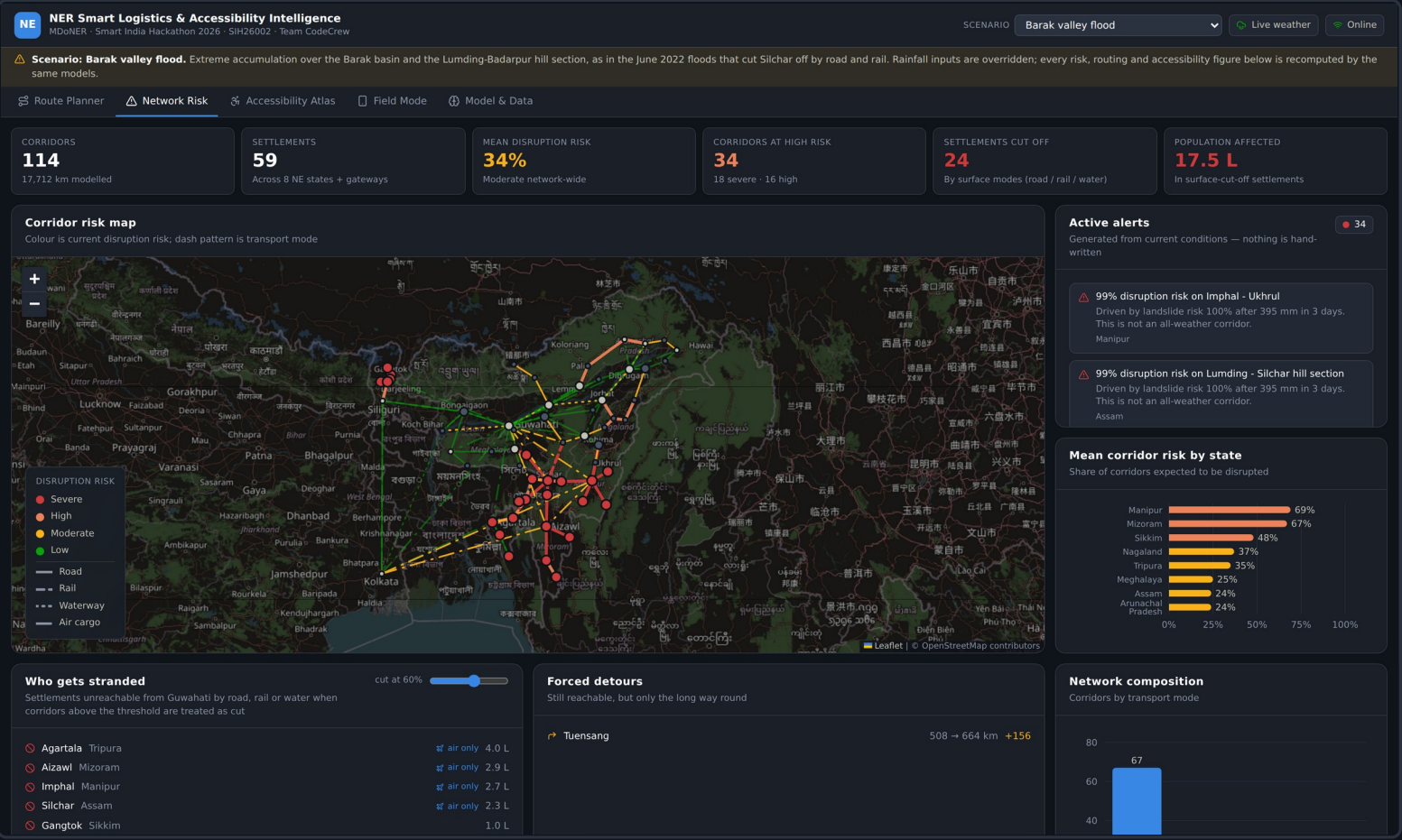
Bridge load class is checked against vehicle gross weight; step-free handling and night travel are consignment inputs.

4 Arrival probability

The product of every link's survival probability, shown per route.

Network Risk

The operating picture: what is at risk right now, and who loses their connection if it closes.



1 Colour is risk, dash is mode

Corridor colour carries current disruption risk; transport mode is carried by line pattern.

2 Alerts are derived

Nothing is hand-written. If the rain stops, the alerts clear themselves.

3 Who gets stranded

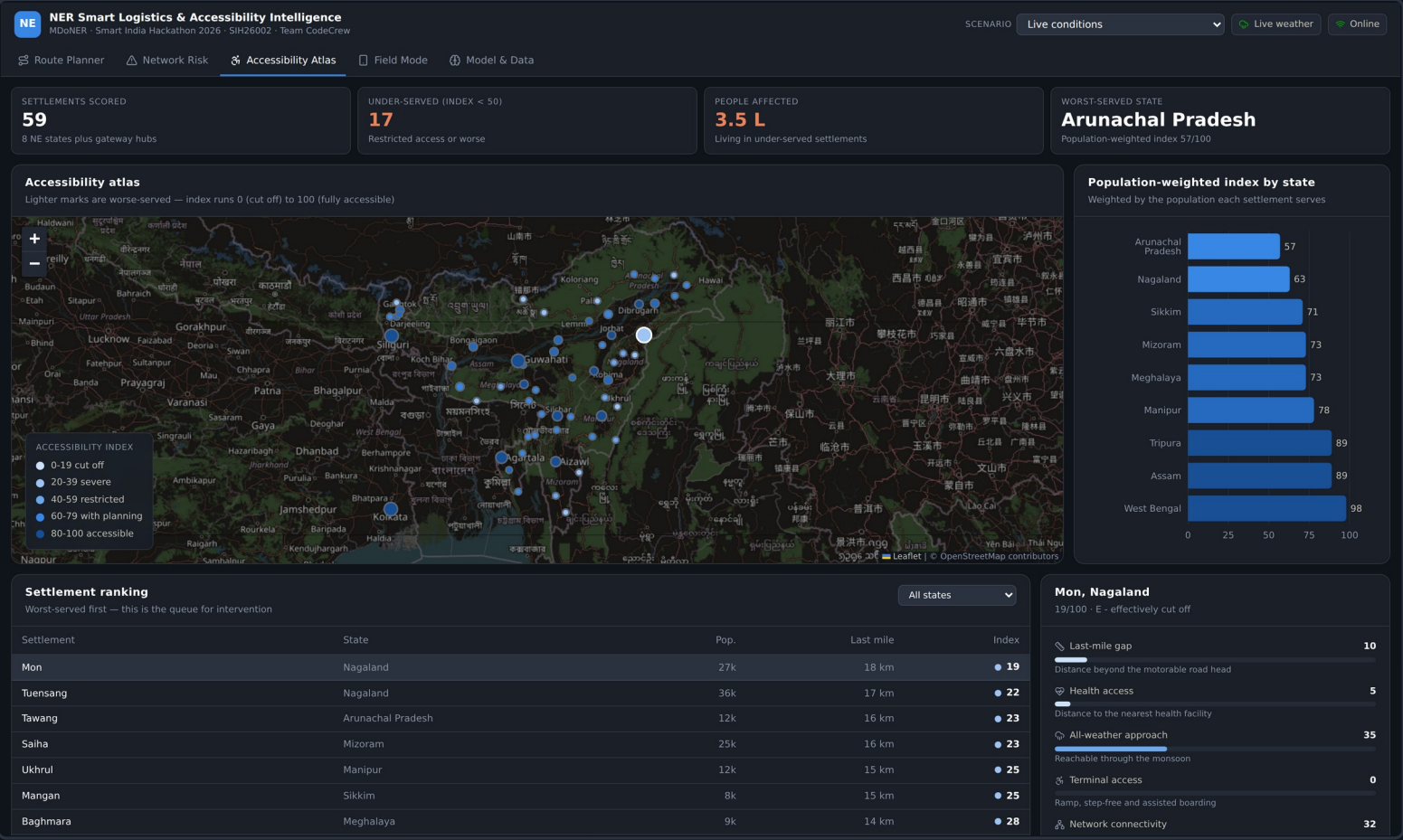
Treat corridors above a threshold as cut, then ask which settlements are unreachable over surface modes.

4 Air excluded on purpose

Relief and resupply move on the ground; an air bridge on paper hides the isolation this view exists to show.

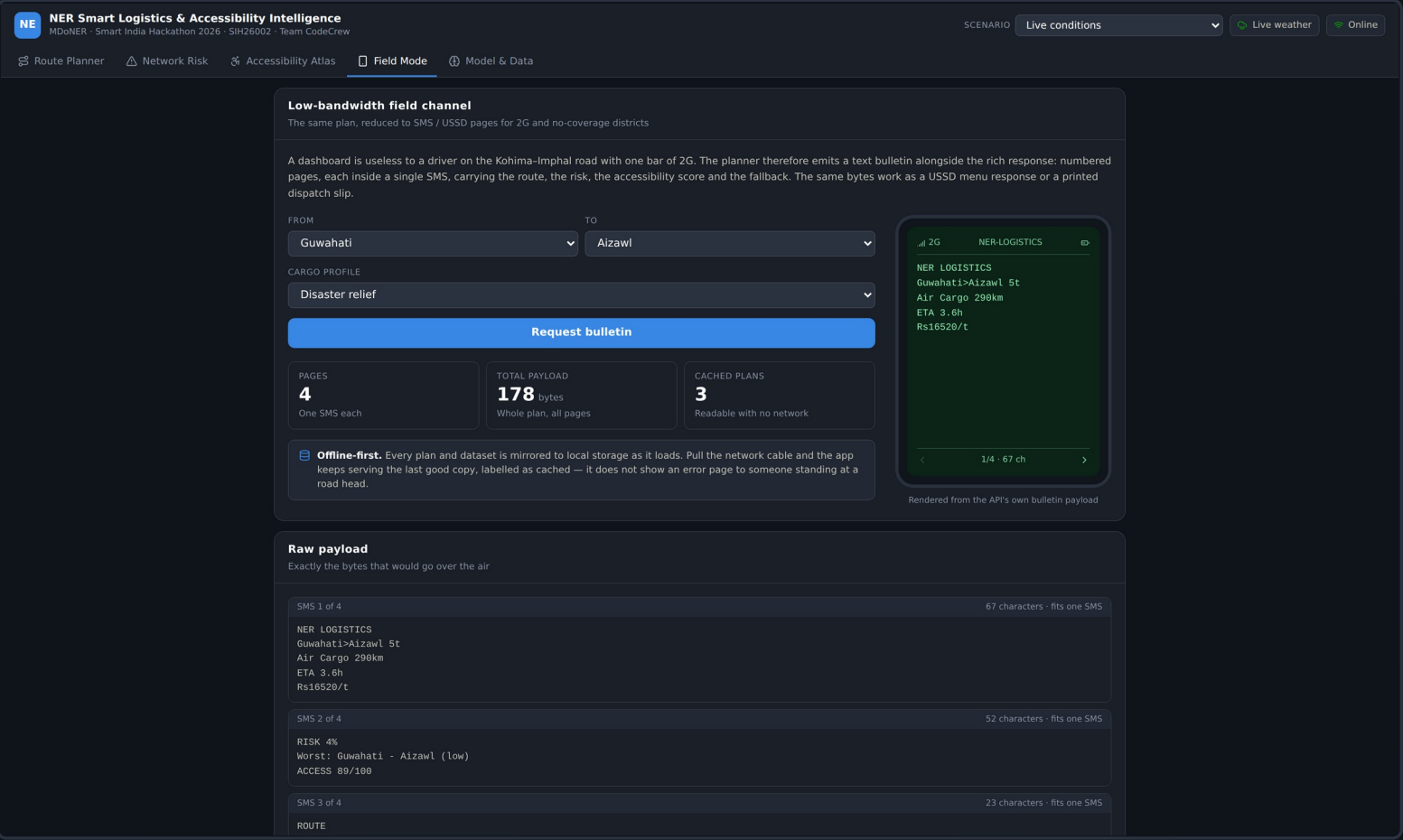
Accessibility Atlas

The differentiator: scoring places, not just routes - because a route that arrives at an unusable road head has not arrived.



Field Mode

A dashboard is useless to a driver on the Kohima-Imphal road with one bar of 2G.



1 Sub-160-character pages

The same plan as numbered SMS pages - route, risk, accessibility and the fallback, each inside one message.

2 One payload, three channels

The same bytes work as an SMS, a USSD menu response, or a printed dispatch slip.

3 Offline-first by default

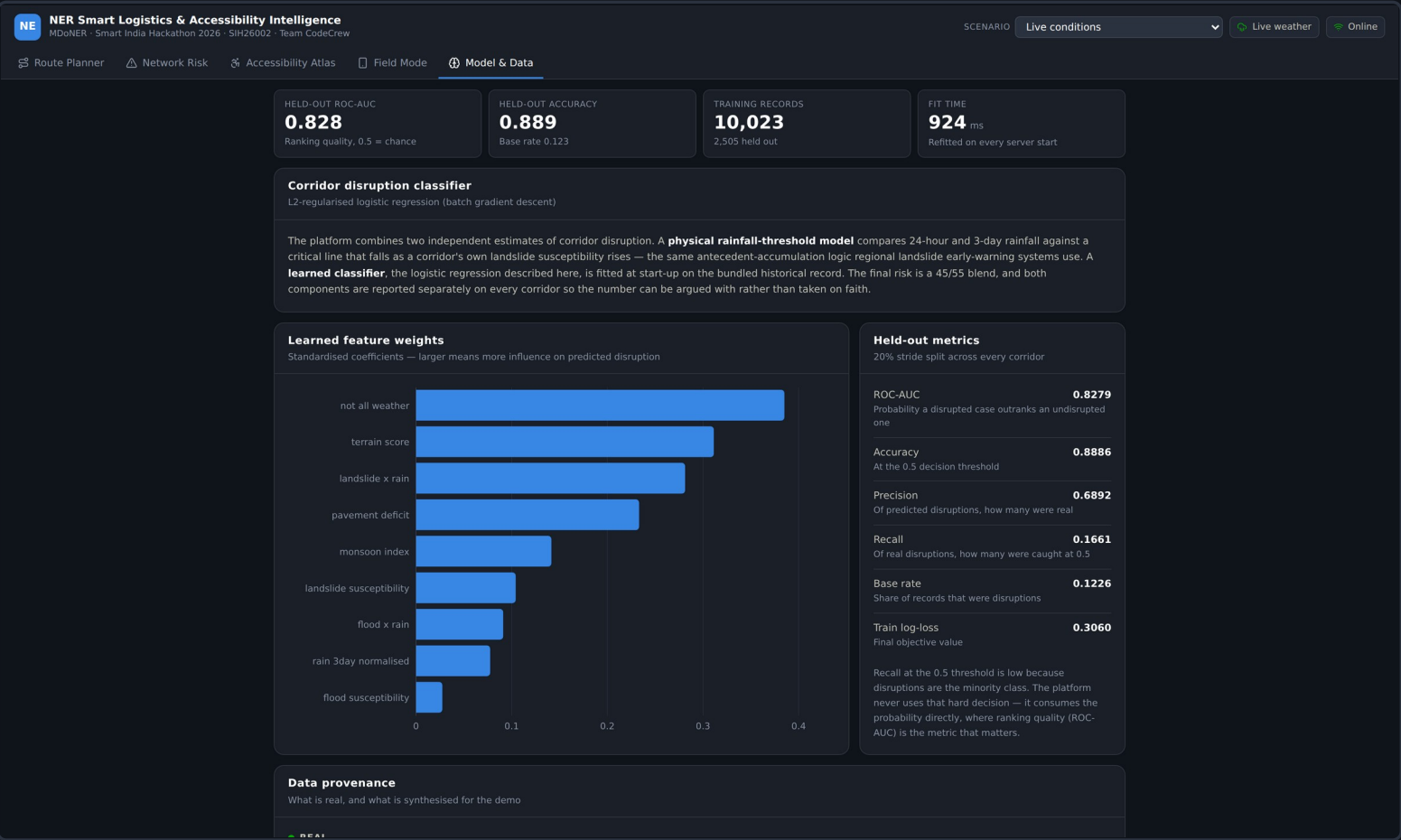
Every response is mirrored to local storage; losing the network shows a labelled cached plan, never an error page.

4 Rendered from the real API

The feature-phone screen shows the actual bulletin payload, with its byte count.

Model & Data

The honesty screen, shipped inside the product rather than kept for the pitch.



1 Learned weights, on screen

The classifier's standardised coefficients, so its reasoning can be inspected.

2 Held-out metrics

ROC-AUC, accuracy, precision, recall and base rate on a 20% split held out across every corridor.

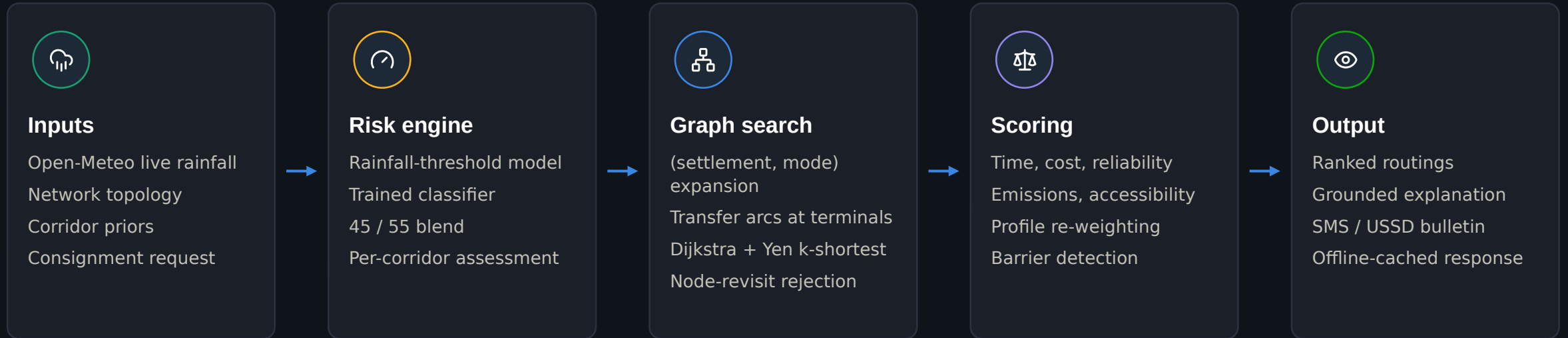
3 Stated provenance

Which data is real and which is the demo layer - in the product, not just the README.

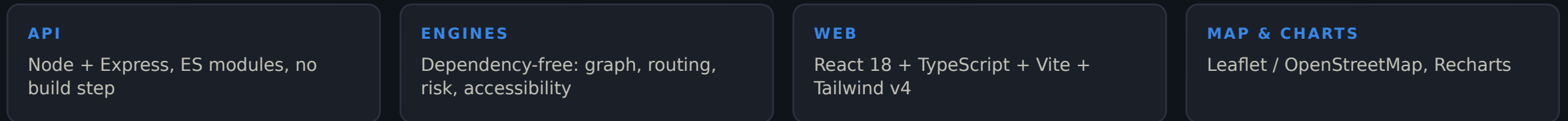
4 Honest about recall

Low recall at the 0.5 threshold is explained: the platform consumes the probability, never the hard decision.

How a request becomes a routing



Stack



No API keys, no database, no external service beyond the free Open-Meteo forecast API - and the platform degrades to monsoon climatology when that is unreachable, so it runs fully offline.

Routing: a graph that knows what a terminal is



The (settlement, mode) expansion

A plain node-to-node graph cannot express transshipment. Switching road to rail at Dimapur costs four hours and a handling fee, and is only possible because Dimapur actually has a rail terminal. So the search space is expanded to (settlement, mode) pairs, and moving between two modes at one place is an explicit transfer arc that exists only where both terminals do.



One objective, five terms

$$\text{cost} = w_{\text{time}} \cdot \text{time} + w_{\text{money}} \cdot \text{cost} + w_{\text{risk}} \cdot (-\ln(1-p)) + w$$

$$+ w_{\text{co2}} \cdot \text{emissions} + w_{\text{access}} \cdot \text{deficit}$$

Reliability enters as $-\ln(1-p)$ because that is the only additive form whose path total corresponds to the real quantity: the probability the whole routing survives is the product of its links' survival probabilities. Terms are normalised to a typical NER movement so no term silently dominates.

Three design decisions that made the answers correct

Yen's k-shortest-paths

Produces genuinely different alternatives - a rail routing, a drier road routing - instead of three cosmetic variants of one corridor.

Node-revisit rejection

Yen is loopless over the expanded graph, so it can legally return "drive Guwahati to Shillong, then fly back through Guwahati". Those paths are rejected.

Impassable gating at 97%

Above that probability a corridor is treated as closed, not merely expensive - otherwise a cheap washed-out road outscores a sound one on price.

The profile changes the answer, not just the label

Same origin, same destination, same 12 tonnes. Only the objective weights differ - and the platform picks a different mode, corridor and cost each time.

Cargo profile	Weighting emphasis	Chosen mode	ETA	Cost	CO ₂
Emergency medical	Time 2.4 · Reliability 1.8	Air cargo	3.5 h	₹ 1,69,440	1,734 kg
Perishable / agri	Time 1.8 · Cost 0.7	Road	18.3 h	₹ 19,834	356 kg
General freight	Balanced across all five	Road	18.3 h	₹ 19,834	356 kg
Bulk freight	Cost 2.6 · Time 0.25	Rail + Road	22.1 h	₹ 17,750	230 kg
Low-emission	Emissions 3.0	Rail + Road	22.1 h	₹ 17,750	230 kg
Disaster relief	Reliability 2.0 · Access 1.6	Air cargo	3.5 h	₹ 1,69,440	1,734 kg

●

Emergency medical flies

Cost weight drops to 0.2, so a 5x price for 15 hours saved is the right trade.

●

Bulk freight takes the train

Cost weight 2.6 against time 0.25 moves the answer onto rail - cheaper and 35% less CO₂.

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Relief buys reliability

Reliability 2.0 and accessibility 1.6 pick the option most likely to actually arrive.

Risk: a physical model and a learned one, reported separately



Physical

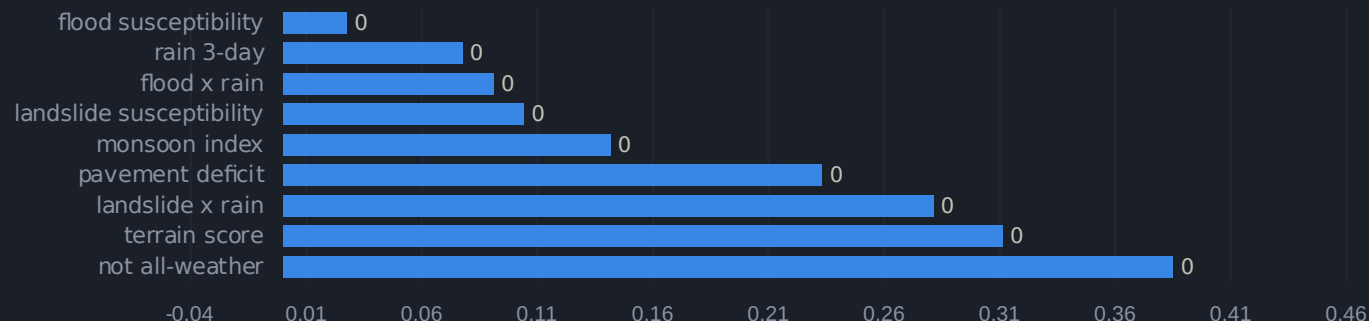
Regional landslide early-warning systems trigger on antecedent accumulation, not the day's intensity. 24-hour and 3-day rainfall are compared against a critical line that falls as the corridor's own susceptibility rises; the exceedance ratio maps through a logistic curve. Flooding uses basin accumulation with a relief factor - water pools on the valley floor and drains above ~600 m.



Learned

An L2-regularised logistic regression, trained by batch gradient descent at every server start on the bundled historical record. It is genuinely fitted, not handed its coefficients - and it recovers the right structure: all-weather status, terrain class and the landslide x rainfall interaction come out as the dominant signals.

Learned feature weights · standardised coefficients



HELD-OUT ROC-AUC

Ranking quality; 0.5 is chance

0.828

HELD-OUT ACCURACY

Base rate 0.123

0.889

TRAINING RECORDS

2,505 held out on a stride split

10,023

FIT TIME

Refitted on every server start

924 ms



Blended 45 / 55

Both components are reported on every corridor, side by side, so the number can be argued with rather than taken on faith. A corridor with no historical signal still gets a sane physical estimate; a corridor the data knows well gets the benefit of it.

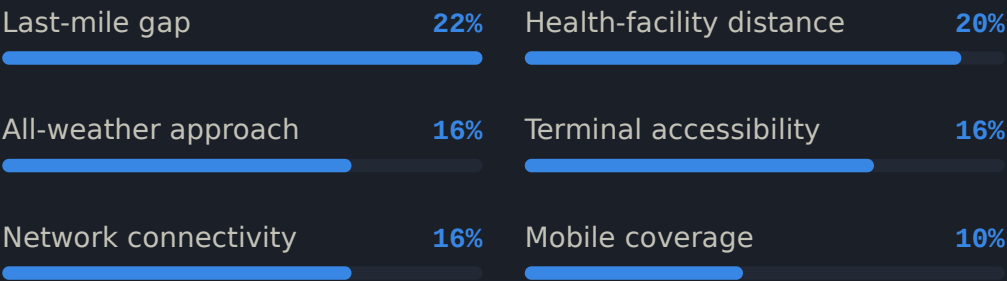
Accessibility: scoring places, not just paths

Two related questions. Is this route physically usable for this consignment and these people? And how well-served is this settlement in the first place?

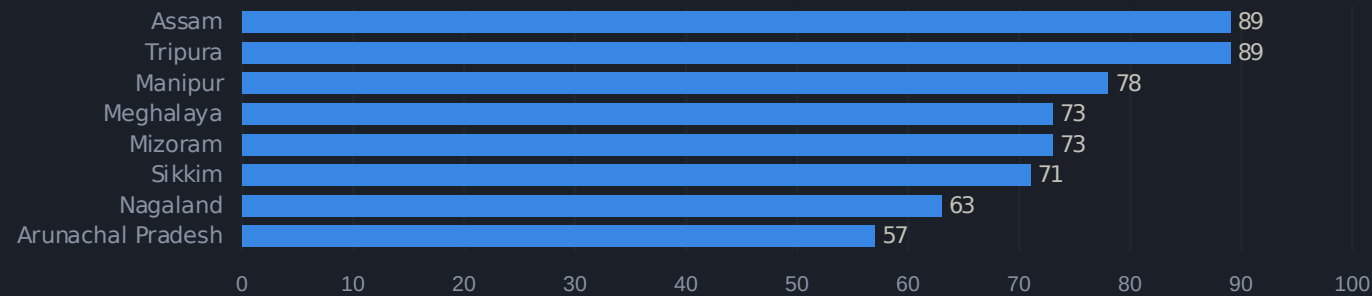
Route accessibility · checked against the consignment

- **Bridge load class** against the vehicle's gross weight - a blocking barrier, with the mitigation stated
- **Gradient and lanes** sustained climbs above 10% restrict heavy and low-power vehicles
- **All-weather status** corridors liable to close under sustained rain are flagged before dispatch
- **Step-free handling** checked at every terminal the load passes through, when required

Settlement index · six sub-scores, 0-100



Population-weighted accessibility index by state



Settlements scored **59**

Under-served (index < 50) **17**

People affected **3.5 L**

Worst served **Mon, 19**

Barak valley flood: the same models, different rainfall

Scenarios substitute rainfall inputs and re-run every model - a genuine what-if, not a scripted demo. This one reproduces the June 2022 floods that cut Silchar off by road and rail.

Live conditions	Barak valley flood	
Mean network risk	21% →	34%
Corridors at severe risk	2 →	18
Active alerts	5 →	34
Settlements surface-cut off	0 →	24
Population affected	0 →	17.5 L

 **Reachable by air only**

Four state capitals and regional hubs lose every surface connection to the Guwahati hub:

Silchar · air bridge only

Agartala · air bridge only

Aizawl · air bridge only

Imphal · air bridge only

This is what actually happened in June 2022. The platform reaches the same conclusion from rainfall alone, and re-routes accordingly: the recommendation for Guwahati to Silchar switches from rail to air cargo, 3.5 hours at 61% arrival probability.

 **And when nothing works, it says so**

Disable air cargo during the flood and the planner does not dress up a bad answer. It reports that every routing crosses a corridor it expects to be closed, shows the least-bad options with their honest 1% and 0% arrival probabilities, and tells the operator to treat dispatch as conditional on ground confirmation.

Where every number comes from



Real

Verifiable against public sources

Settlements	59 places - coordinates, elevation, state, terminal facilities
Road corridors	NH-27, NH-2, NH-6, NH-10, NH-13, NH-306, NH-54 and others, by number and routing
Rail	NF Railway network, including the Lumding-Badarpur hill section
Waterways	NW-2 Brahmaputra and NW-16 Barak
Air	Scheduled sectors plus UDAN-RCS routes
Weather	Open-Meteo current + forecast precipitation, 15-minute refresh
Mode economics	Indicative freight rates, speeds, CO ₂ per tonne-km



Demo layer

Plausible and internally consistent, not surveyed

Corridor priors	Landslide and flood susceptibility per corridor
Road condition	Pavement index, bridge load class, gradient, lane class
Night safety	Whether night travel is advised on a corridor
Settlement access	Ramp, step-free, assisted boarding
Service distance	Last-mile gap and nearest health facility
Coverage	Mobile data availability per settlement
Training history	Synthesised disruption records - see the note below



One file stands between the demo and real data

Open per-corridor incident records for the NER are not publicly downloadable, so the classifier is fitted on a synthesised history generated from NE-India monsoon climatology and the corridor priors. Swapping in real NHAI / NDMA / state-PWD incident data means replacing `server/src/data/history.js` - nothing else changes.

Built to be checked, not just demonstrated

END-TO-END CHECKS

25

Run with `npm run smoke` - no mocks, real engines

ENGINE DEPENDENCIES

0

Graph, routing, risk and accessibility are plain code

NETWORK VALIDATION

boot

A malformed corridor fails loudly at start-up

MODEL RETRAIN

boot

Refitted every start; metrics published at `/api/model`

Three defects the tests and review caught - and what each one was



Routes that teleported

Yen's algorithm is loopless over the expanded (settlement, mode) graph, where Guwahati-road and Guwahati-air are different vertices. It legally returned "drive Guwahati to Shillong, then fly Shillong-Guwahati-Imphal" - a path that revisits Guwahati and is nonsense on the ground.

FIX Over-fetch candidates and reject any path that revisits a physical settlement.



A 1% route recommended

Under the flood scenario the planner recommended a rail line with a 1% chance of arriving, because the $-\ln(1-p)$ risk term saturates at the probability cap and a cheap corridor won on price.

FIX Corridors above 97% disruption are treated as closed. If that leaves no route, the planner re-runs ungated and flags the answer as least-bad.



Totals that did not add up

Origin loading and destination hand-over were counted in the quoted total but never shown as legs, so an operator reading the itinerary could not reconcile the cost.

FIX Handling is emitted as itinerary legs; a test now asserts the legs sum to the total.

What production would need

Phase 1

0-3 months

Real data substitution

- NHAI / NDMA / state-PWD incident records replace the synthesised training history
- PM Gati Shakti and NESIDS layers for corridor and asset attributes
- State transport department feeds for bridge load class and pavement condition
- Retrain and re-validate the classifier on real outcomes

Phase 2

3-9 months

Pilot in one state

- Modular, state-by-state rollout - start with one accessible state, not all eight
- Community operators collecting local-language accessibility data
- SMS / USSD gateway integration for the field channel
- Operator feedback loop on route outcomes, which is the data RL would need

Phase 3

9-24 months

Scale and learn

- Satellite and IoT road-sensor ingestion for near-real-time corridor state
- Reinforcement learning over accumulated outcome data - once there is data to learn from
- Multilingual interfaces; security handling for border districts
- Integration with UDAN cargo scheduling and inland-waterway berth planning

The architecture already assumes this path: the demo data layer is isolated in three files, and the engines that consume it do not change when it is replaced.

Run it yourself

The whole platform starts with two commands and needs no API key, no database and no external service beyond a free weather API it can also do without.

```
$ npm install
```

```
$ npm run dev
```

 web on :5173, API on :5174

```
$ npm run smoke
```

 25 end-to-end checks, no server needed

5 product surfaces

114 corridors modelled

59 settlements scored

25 end-to-end checks

Team CodeCrew

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